Building Block Design for Stacked Polyphase Bridge Converters for High-Voltage Traction Drives

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***Abstract*—In a stacked polyphase bridge (SPB) converter the battery voltage is shared by series-connected three-phase submodules, so the transistor voltage class is a free design variable. This paper designs the phase-leg building block of a 1200 V, 300 kW SPB traction inverter. Since the phase current is independent of the number of submodules, low-voltage GaN transistors are favored, and the turn-off overshoot at a realistic commutation-loop inductance sets 16 submodules of 75 V with 100 V devices. Four EPC2361 per switch, driven as two mirrored pairs by one dual-channel isolated driver each, keep the junction temperature at 69 °C and the peak drain–source voltage below 90 V. The dc link of a three-phase level is sized for the voltage ripple with the dc-bias derating of class-II MLCCs. The leg is built in a double-pulse-test version with a coaxial shunt per cell and a load-test version without, and three legs share a laminated PCB busbar. Electromagnetic extraction of the routed boards gives a commutation-loop inductance of 0.41 nH per cell without and 0.65 nH with the shunt, equal in the four parallel cells within 1.8 %. A double-pulse simulation with the extracted parasitic network confirms equal current sharing among the cells and a peak voltage of 86 V at the peak phase current.**

***Index Terms*—GaN, paralleling, parasitic inductance, stacked polyphase bridge, traction inverter.**

I. Introduction

Traction batteries of heavy-duty electric vehicles are moving to 800 V and above, which pushes two-level inverters to 1200 V transistors with large voltage steps at the machine terminals. The stacked polyphase bridge (SPB) converter [1], [2] connects N three-phase bridges in series on the dc side, each feeding one winding set of a multiphase machine. Every submodule blocks only V_{bus}/N, which allows low-voltage transistors, reduces the voltage step at the windings and the dc-link capacitance [3], [4], and allows a faulty submodule to be bypassed. Control [2], [5], dc-link stability [6], machine design [7], [8] and integration [9] have been studied, and a GaN-based SPB has been found competitive with a three-level flying-capacitor inverter [10].

The submodule itself, however, has received little attention. At 1200 V and 300 kW each leg conducts more than 200 $\mathrm{A_{rms}}$ at 75 V, so several GaN transistors must be paralleled. Paralleled low-voltage GaN stages are limited by turn-off overshoot [11], by uneven dynamic current sharing [12], [13] and by gate disturbances [14]; symmetric layouts with up to eight devices per switch have been demonstrated at 48–120 V [15], [16]. In those designs the bus voltage is given. In an SPB it is not, and the submodule voltage, the device count and the layout are coupled through the loop inductance.

This paper derives the SPB building block from the inverter specification and verifies the resulting hardware. The contributions are: 1) a selection of the device voltage class and the submodule voltage that accounts for the commutation-loop inductance; 2) a gate-drive arrangement for four paralleled GaN transistors per switch based on a current-sharing sensitivity analysis; 3) a four-cell leg board, built in a test and an operating version, and its verification by Q3D extraction of the routed copper and by a double-pulse simulation with the complete extracted parasitic network.

II. Building Block Requirements

For N submodules sharing the power equally (Fig. 1a), one leg delivers $P_{\mathrm{leg}} = MV_m I_{\mathrm{ph}} \cos\varphi/(2\sqrt{2})$, so the inverter power is

$$P = 3NP_{\mathrm{leg}} = \frac{3M\,V_{\mathrm{bus}}}{2\sqrt{2}}\, I_{\mathrm{ph}} \cos\varphi. \tag{1}$$

The phase current does not depend on N: with $M = 1.15$ and $\cos\varphi = 0.9$, 300 kW at 1200 V require 228 $\mathrm{A_{rms}}$ (322 A peak) in every leg of every submodule. The number of submodules therefore only scales the voltage, and with it the number of transistors and drivers. The peak drain–source voltage is limited to 90 V for 100 V devices, and the switching frequency is set to 50 kHz to keep the submodule capacitors small.

III. Device and Submodule Voltage

Since every submodule carries the full phase current, the conduction loss of the inverter scales with $NI^2R_{\mathrm{DS(on)}}/n$ for n devices per switch, and the output-charge loss with $Nnf_{\mathrm{sw}}C_{\mathrm{oss,tr}}V_m^2$. Optimizing n gives the minimum loss

$$P_{\min} = 6\,I\,V_{\mathrm{bus}}\sqrt{f_{\mathrm{sw}}\,R_{\mathrm{DS(on)}}C_{\mathrm{oss,tr}}}, \tag{2}$$

in which the voltage class cancels to first order. Devices are thus ranked by the time constant $\tau = R_{\mathrm{DS(on)}}C_{\mathrm{oss,tr}}$ and, because conduction dominates at this current, by the on-resistance per volt of blocking capability. Among ten GaN transistors of the 100–650 V classes, the 100 V parts lead in both (Fig. 2a,b); EPC2361 (0.75 mΩ, 3 × 5 mm) has the lowest τ and 7.5 μΩ/V, three to thirty times below the higher-voltage parts. The price is a larger number of submodules (Fig. 2c), which the modular SPB accommodates. The Kelvin source

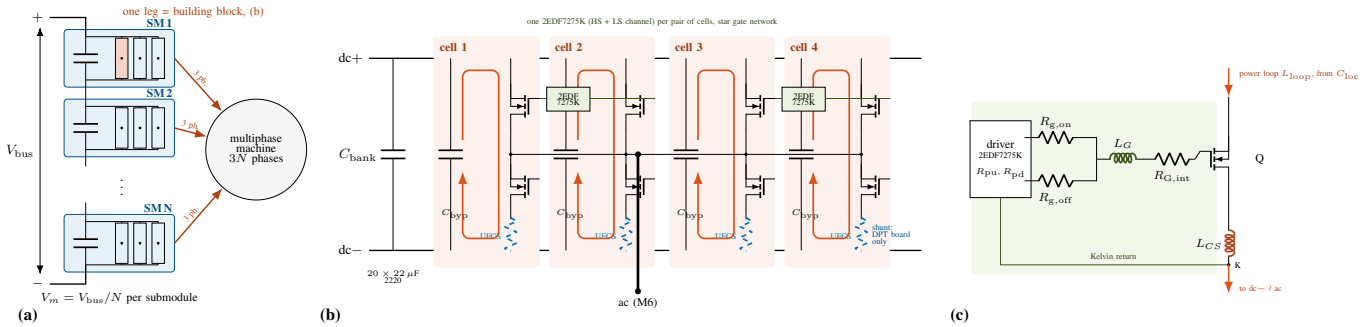


Fig. 1. (a) SPB inverter with N submodules (SM) of $V_m = V_{\rm bus}/N$. (b) Building block, one phase leg: four commutation cells (QH, QL: EPC2361) with local bypass capacitors $C_{\rm byp}$, a common capacitor bank $C_{\rm bank}$ and one ac output; one dual-channel driver serves each pair of cells. The shunt (dashed) is fitted on the test board only. (c) Gate circuit of one transistor with the common-source inductance L_{CS} up to the Kelvin pad K.

pad of the EPC2361 keeps the source inductance out of the gate loop (Fig. 1c).

The submodule voltage follows from the overshoot at turn-off, $V_m + L_{\rm loop}\, di/dt \le 90$ V. A vertical loop over a 0.1 mm dielectric reaches about 0.5 nH per cell, whereas a measurement shunt raises it to about 1 nH. With the turn-on rate of a 2EDF7275K-driven EPC2361 ($R_{g,\rm on} = 2\,\Omega$), 33 A/ns at 80 A per device, the admissible submodule voltage is 73 V at 0.5 nH and 57 V at 1 nH (Fig. 4).

An even N is needed for symmetric winding sets, so $N = 16$ with $V_m = 75$ V is chosen. This requires a loop close to 0.5 nH in the operating hardware; $N = 18$ would add 12.5 % transistors and conduction loss.

IV. Devices per Switch and Gate Drive

A. Number of paralleled devices

Double-pulse simulations of one EPC2361 per switch (LTspice, 75 V) bound the current per device. At 120 A the overshoot reaches 102 V with 0.5 nH, and at 80 A already 93 V with 1 nH, so the current per device is limited to about 80 A at the peak. A larger loop lowers the turn-on energy (21.8 versus 28.3 µJ at 80 A for 1 and 0.5 nH), because it takes $L_{\rm loop} di/dt$ of the voltage during the current rise, but it raises the overshoot and the turn-off energy.

The leg loss over a fundamental period is computed from these energies, with the junction temperature and $R_{\rm DS(on)}$ solved together for the thermal path of the load-test hardware: a 50 mm fan heatsink (0.65 K/W) on an aluminium spreader and, per device, 7.6 K/W from junction to heatsink, dominated by the insulating gap pad (5.3 K/W). Fig. 3 shows the result. One device per switch runs away thermally and two reach 141 °C. Three keep T_j at 84 °C but exceed 90 V. Four devices dissipate 18.0 W per leg (12.5 W conduction) at 69 °C and 40 °C ambient, with 83 V peak voltage at 0.5 nH but 93 V at 1 nH. More devices save less than 3 W per leg but add drivers. Four devices per switch are chosen, which also form two equal pairs for the gate drive below.

B. Gate drive

Static sharing is benign because $R_{\rm DS(on)}$ rises with temperature; dynamic sharing is not. Fig. 5 and Table I show the imbalance of the switching energy for one asymmetry at a time. A 6 nH longer gate loop delays the second device by about half the current rise time and shifts most of the turn-on energy to the first. Gate-loop and common-source mismatches dominate, while a power-loop mismatch of 0.3 nH is nearly harmless. The gate paths must therefore be equal by construction. Driving all four devices of a switch from one channel is not an option: the slower gate charging raises the turn-on energy by 31 % and the peak sink current exceeds the 8 A rating. Instead, one 2EDF7275K (4/8 A, isolated high- and low-side channel, 3 ns channel mismatch) is placed on the symmetry line between two mirrored cells and drives both gates through a star with Kelvin returns (Fig. 1b). Each supply is split by a Zener diode into $+5.1$ V and -1.2 V. The negative bias absorbs the Miller-induced gate bump of the off device, $\Delta v_{\rm GS} \approx Q_{\rm GD}/C_{\rm GS} \approx 1.06$ V, which is otherwise close to the 0.8 V minimum threshold.

TABLE I
Switching-Energy Imbalance of Two Paralleled EPC2361 (LTspice, 75 V, 120 A)

Asymmetry	Imbalance	Layout measure
Power loop, L_D +0.3 nH	2 %	mirrored cells
Gate loop, L_G 2 vs. 8 nH	19 %	star gate network
No Kelvin, L_{CS} +0.2 nH	29 %	Kelvin return per device
Threshold, $V_{\rm th}$ +0.3 V	7 %	one channel per pair

V. DC Link of a Three-Phase Level

The three legs of a submodule share one dc bus, so the low-frequency ripple cancels and only the switching-frequency component remains. A switching-function model with space-vector modulation gives 85 $A_{\rm rms}$ capacitor current per level and a charge excursion of 860 µC per switching period at 50 kHz. A ripple of 3 % (2.25 V) therefore requires 127 µF effective capacitance per leg board; the voltage ripple, not the current, sizes the dc link. At 75 V, 100 V class-II MLCCs retain only 15–29 % of their nominal capacitance. The 22 μF 2220 part with the highest retention (29 %, 6.35 μF) is selected, and twenty of them are placed on each leg board (Fig. 7); the ripple current alone would need only 7 parts.

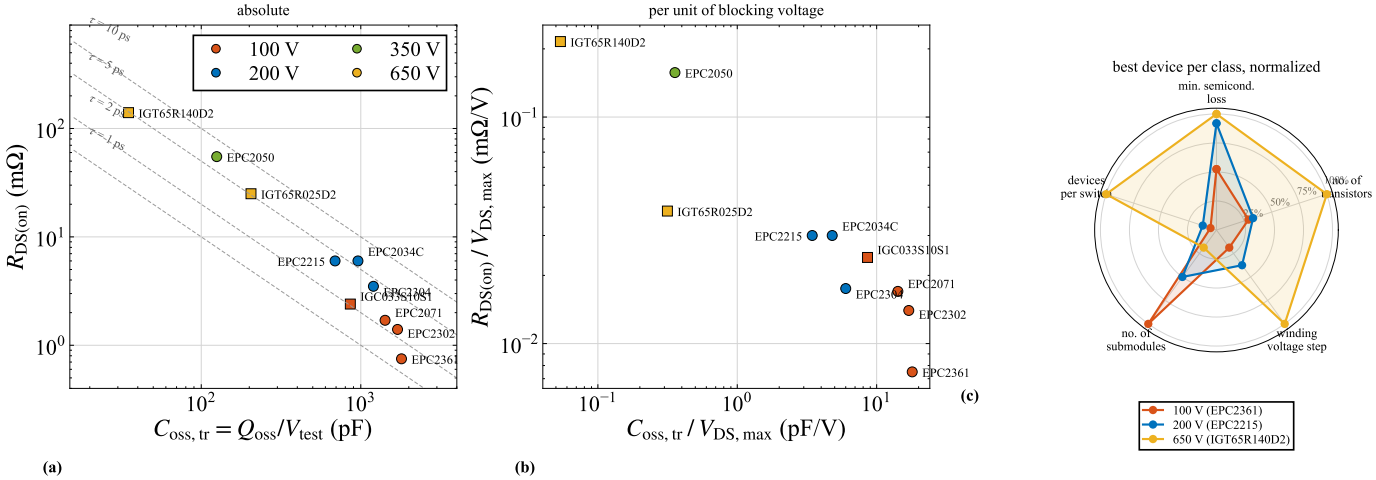


Fig. 2. Device selection for a 1200 V SPB (datasheet values). (a) $R_{\mathrm{DS(on)}}$ against $C_{\mathrm{oss,tr}}$ with lines of constant τ; (b) both per volt of blocking capability; (c) SPBs built from the best device of each class, normalized: minimum loss (2), devices per switch, submodules, voltage step and transistor count.

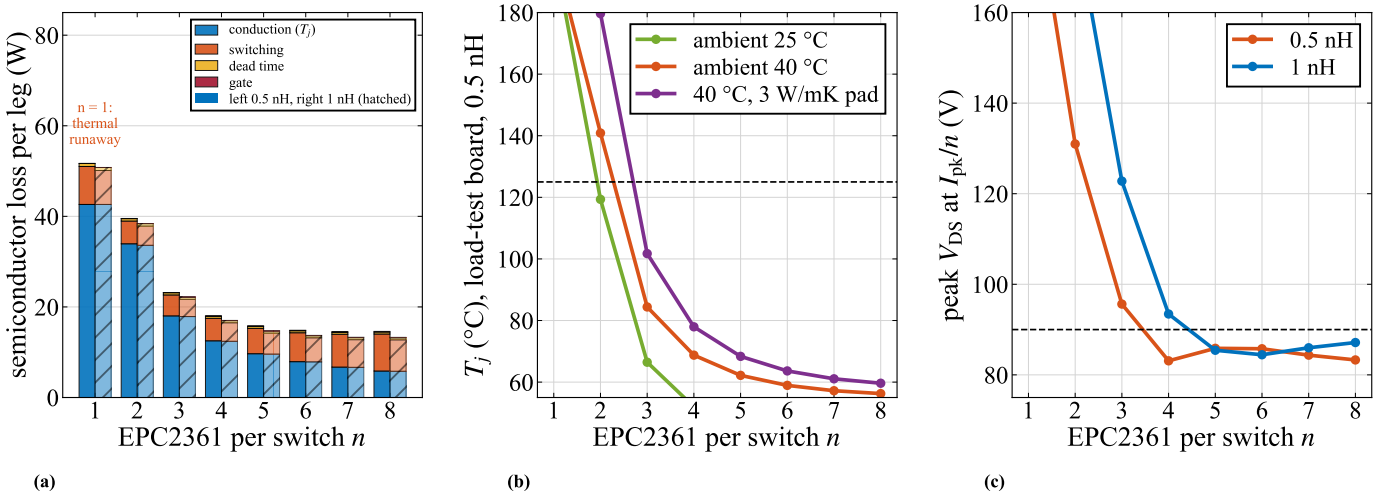


Fig. 3. n EPC2361 per switch at 228 A_{rms} and 50 kHz. (a) Semiconductor loss per leg, 0.5 nH (left) and 1 nH (right, hatched); (b) junction temperature on the load-test heatsink; (c) peak V_{DS} at the peak current per device.

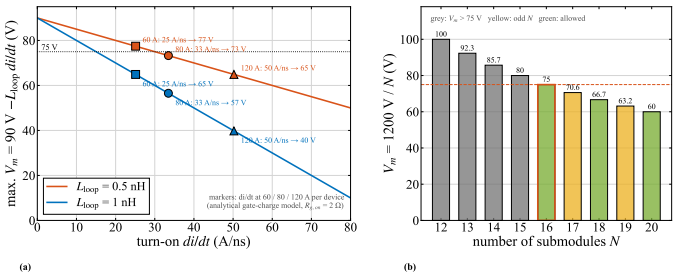


Fig. 4. (a) Admissible submodule voltage $90\,\mathrm{V} - L_{\mathrm{loop}}\,di/dt$ for 0.5 and 1 nH; markers: turn-on di/dt at 60, 80 and 120 A per device. (b) $V_m = 1200\,\mathrm{V}/N$; green: admissible and even.

Ten 1 μF 0805 capacitors per cell, on both board sides at the high-side drains, close the commutation loop.

VI. Leg Board and Verification

The four-layer leg board (2 oz, 0.1 mm dielectric between L1 and L2) carries the four cells as two mirrored pairs, each with its driver in the centre (Fig. 6). Each commutation loop is vertical: the forward path on L1 returns on the dc− plane on L2 directly below. One dc terminal pair feeds a laminated row across the board that supplies both the capacitor bank and all cells, and the four switch nodes are joined in copper and leave through a single bolted ac terminal. Symmetry required two measures. The interleaved drain and source stripes of the EPC2361 leave no room for a via row, so the low-side source stripes use vias in the pad, and any via removed for clearance is removed together with its mirror image.

A shunt in every low-side source is needed to measure the cell currents but lengthens the loop and blocks the heatsink. The board is therefore built in two versions from the same design. The double-pulse-test (DPT) board carries a 1 mΩ

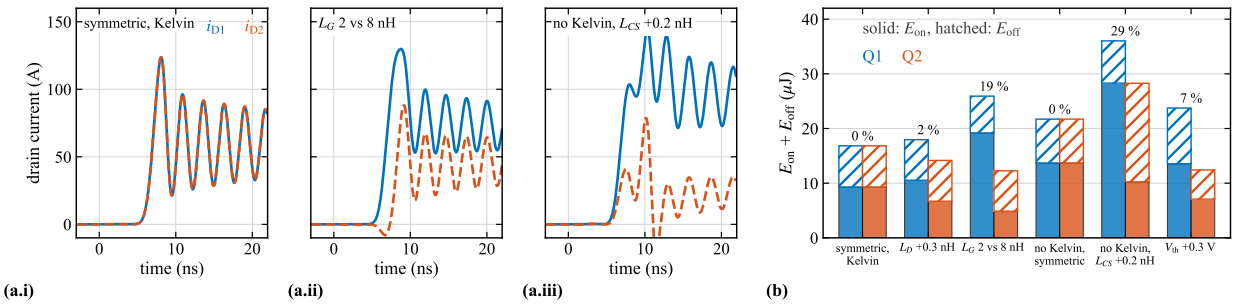


Fig. 5. Current sharing of two paralleled EPC2361 (LTspice, 75 V, 120 A). (a) Drain currents at turn-on: (i) symmetric with Kelvin connections, (ii) gate loops of 2 and 8 nH, (iii) no Kelvin connection and 0.2 nH common-source mismatch. (b) Turn-on (solid) and turn-off (hatched) energy of both devices.

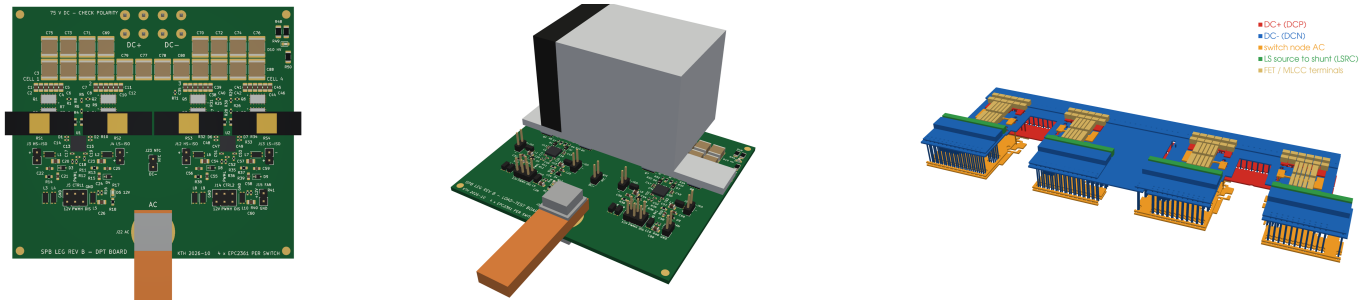


Fig. 6. Leg board, four EPC2361 per switch (89.5 × 80 mm, four layers). (a) Double-pulse-test board with a coaxial shunt per cell; (b) load-test board with fan heatsink; (c) Q3D model of the cells of the test board, generated from the routed copper (red dc+, blue dc−, orange switch node; z stretched).

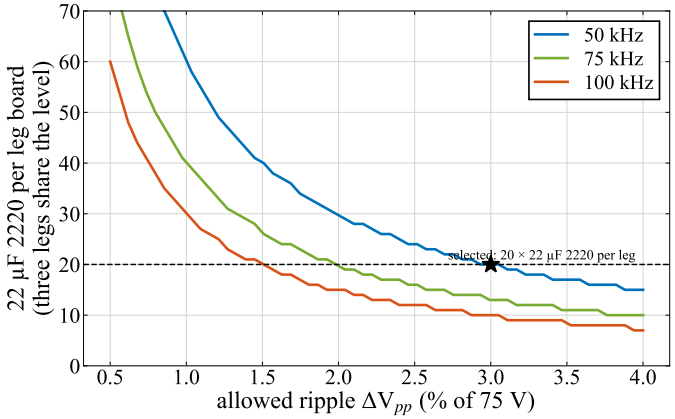


Fig. 7. Number of 22 μF 2220 MLCCs per leg board against the peak-to-peak ripple of the 75 V level, including the dc-bias derating; star: selected design.

TABLE II
Q3D Results of the Two Boards

	DPT	Load test
L_{loop}, cell 1/2/3/4 [nH]	0.644/0.646/ 0.642/0.654	0.405/0.406/ 0.407/0.410
Copper / bypass ESL / shunt [nH]	0.46/0.06/0.12	0.34/0.06/–
$\hat{v}_{DS}$ at 30 A/ns [V]	94	87
Static imbalance, HS/LS [%]	TBD/TBD	TBD/TBD
Board copper loss [W]	TBD	TBD

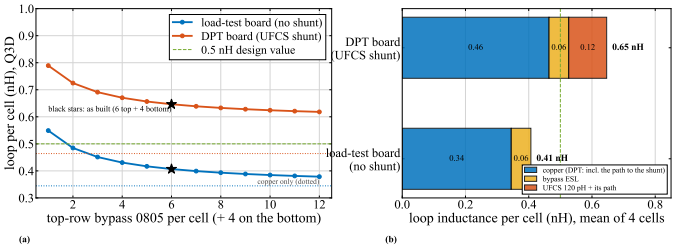


Fig. 8. Loop inductance per cell (Q3D). (a) Against the number of top-side bypass capacitors (stars: as built); (b) contributions of copper, capacitor ESL and shunt.

coaxial shunt (PMK UFCS) per cell on a shortened land and needs no heatsink. The load-test board connects the low-side sources through blind vias to the dc− plane and carries the heatsink.

The routed boards were verified in Ansys Q3D with the loop evaluated with all four cells switching. To keep the solution time short, each quantity is extracted from the smallest model that contains it: the cell band at 100 MHz for the loop and the full board at dc for the resistances. Table II and Fig. 8 summarize the results. The load-test board reaches 0.41 nH per cell, of which 0.34 nH is copper and 0.06 nH capacitor ESL, and the four cells agree within 1.3 %. It thus meets the 0.5 nH target and gives about 87 V peak at 30 A/ns. On the DPT board the shunt and its connection add 0.24 nH, which costs overshoot margin at full switching speed (Section VII)

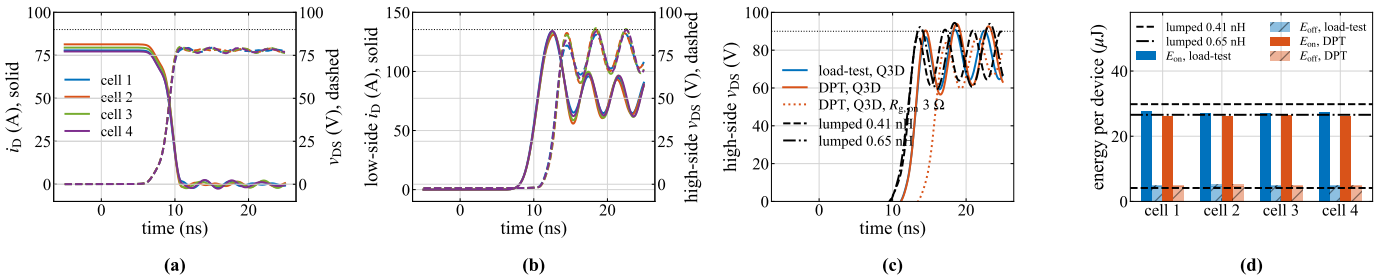


Fig. 9. Double-pulse simulation with the Q3D-extracted copper, 75 V, 316 A load current (≈ 80 A per device), $R_{g,on} = 2\,\Omega$. (a) Low-side turn-off and (b) low-side turn-on on the load-test board, all four cells: drain current (solid) and v_{DS} (dashed). (c) High-side v_{DS} during the low-side turn-on, cell with the highest peak, compared with the lumped model at the extracted loop inductance. (d) Switching energy per device in each cell; lines: lumped model.

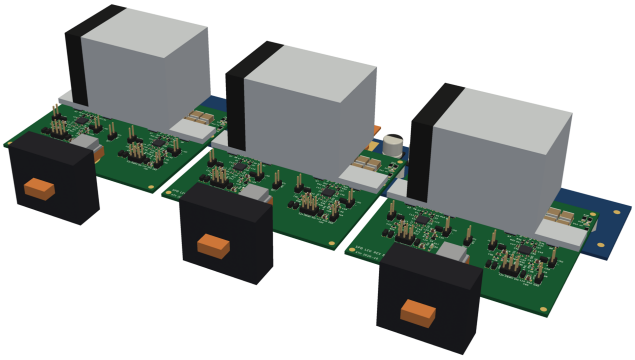


Fig. 10. One 75 V level: three load-test boards on the laminated PCB busbar, ac bars through Hall-effect current transducers.

TABLE III
COMPARISON WITH PARALLELED-GAN HALF BRIDGES

Work	Devices per switch	Operating point	L_{loop}
[12]	4 × 650 V	400 V, 240 A (DPT)	0.7 nH
[13]	4 × 100 V	48 V, 328 A (DPT)	1.14 nH
[15]	8 × 100 V	48 V, 125 A_{rms}	–
[16]	8 × 200 V	120 V, 155 A	0.34 nH*
This work	4 × 100 V	75 V, 228 A_{rms}	0.41 nH/cell

*analytical estimate

and confirms that the shunt does not belong in the operating hardware. Beyond four bypass capacitors per cell the loop hardly decreases (Fig. 8a); the remaining capacitors provide capacitance rather than lower inductance.

Three load-test boards form a level on a four-layer laminated PCB busbar (Fig. 10). The busbar only exchanges the switching-frequency ripple between the capacitor banks of the three legs. The inductance between neighbouring legs is 1.46 nH (1.98 nH between the outer legs at 1 MHz), small against the reactance of a bank at 50 kHz, so the banks share the ripple. The 94.5 mm leg pitch also limits the heatsink to the 50 mm size. Sixteen such levels in series form the 1200 V inverter.

Table III places the design among paralleled-GaN half bridges. With four parallel cells, each with its own loop of 0.41 nH, the leg reaches a lower loop inductance than the module-type layouts [12], [13], while the per-pair gate drive addresses the sharing mechanisms reported for them. The remaining limits are the overshoot margin, which depends on the turn-on speed and would require a regulated submodule voltage if the battery voltage rises well above nominal, and the thermal path, which is dominated by the insulating gap pad and would be replaced by a ceramic-insulated path in an integrated design [13].

VII. VALIDATION WITH THE EXTRACTED PARASITICS

The device-level design of Sections III and IV relies on a lumped loop. A routed board adds what this model hides: the four cells reach the capacitor bank and the ac terminal through different copper, couple to each other, and see their bypass capacitors at different distances. To check the design against these effects, the extracted copper was inserted into the double-pulse circuit of Section IV. The partial-inductance matrix of the cells, bypass pads, shunt lands and switch-node copper is represented by one inductor per terminal with every mutual inductance as a coupling coefficient (29 inductors and 400 couplings for the load-test board). The dc resistance matrix, whose mutual terms describe the shared copper, is applied through behavioral sources. Eight EPC2361 models, the bypass capacitors with their dc-bias capacitance and ESL, the shunts, the capacitor bank and two drivers with star gate networks and Kelvin returns complete the circuit. The load current is switched at the peak phase current.

Fig. 9 shows the result. The four cells carry 77–81 A at turn-off and their turn-on energies differ by 2 % (0 % on the DPT board), so the mirrored layout and the per-pair gate drive leave none of the sharing mechanisms of Table I active. The low-side turn-off is uncritical (81 V), because the device limits its current fall to about 40 A/ns. The critical event is the low-side turn-on, at which the high-side device is charged at 46 A/ns. Its drain–source voltage peaks at 91 V on the load-test board, in agreement with the lumped model at the extracted loop (91 V), and at 94 V on the DPT board. The energies agree with the lumped model within 10 % for turn-on (27.3 versus 29.8 µJ) and within 1 µJ for turn-off. The peak gate bump of the high-side devices, 1.2 V with a 0 V off-state, exceeds the 0.8 V minimum threshold and confirms the need for the negative off-state voltage of Section IV.

The simulation therefore validates the lumped design but shows that the load-test board operates exactly at the 90 V limit with $R_{\mathrm{g,on}} = 2\,\Omega$. With $3\,\Omega$ the turn-on slows to 34 A/ns and the peak drops to 86 V on the load-test board and 87 V on the DPT board. The turn-on energy rises by 20 % (32.8 µJ per device), which adds less than 1 W per leg. Both boards are therefore operated with $R_{\mathrm{g,on}} = 3\,\Omega$.

VIII. Conclusion

Because the phase current of an SPB does not depend on the number of submodules, the device and the submodule voltage can be chosen for the lowest loss and the available overshoot margin. For a 1200 V, 300 kW inverter this leads to 16 submodules of 75 V with four 100 V GaN transistors per switch, driven in mirrored pairs by one isolated dual-channel driver each. Q3D extraction of the routed board gives 0.41 nH per cell and equal loops in all four cells. A double-pulse simulation with the full extracted parasitic network confirms the lumped design: the cells share current within 3 % and the switching energy within 2 %. It also shows that the turn-on has to be slowed slightly ($R_{\mathrm{g,on}} = 3\,\Omega$) to keep the peak voltage at 86 V; the test version with per-cell shunts (0.65 nH) then stays at 87 V. Double-pulse and continuous tests of a three-phase level are the next step.

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